

Simulating the light output of SLIMER using Geant4

Tristan Allison

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1 Introduction

SLIMER (Scintillator-Layered Imaging Microscope for Environmental Research) is a system comprised of a microscope, EMCCD camera, and a scintillator used to detect events from radioactive sources.[2] The system originally developed at Los Alamos National Laboratory for soil analysis, but is now used at Tennessee Tech for student research.[1]

SLIMER is used for detecting radioactive decay events by converting particle interactions into photons. The scintillator that is utilized in SLIMER is made of CsI(Tl). [3] A radioactive sample is placed in front of the scintillator, so that when the sample decays, the resulting particle interacts with the scintillator.[1] When such events happen, light is produced by the scintillator and travels through the objective and microscope to the EMCCD camera.

Within the EMCCD camera, the signal is amplified through an electron-multiplication process. Electrons generated in the sensor are passed through a gain register where impact ionization produces secondary electrons, significantly amplifying the original signal.[2] This process allows the system to detect extremely faint light produced by individual decay events.

However, the data collected by SLIMER has a lot of noise, making difficult to discriminate events from background noise. This project aims to develop a simulation of SLIMER to better understand detector behavior and identify sources of signal loss. The simulation will be implemented using a particle-tracking simulation toolkit, Geant4. Geant4 has the capability to simulate motion of particles through materials and other physical interactions. By replicating the interactions within the detector apparatus, the simulation will generate data that can be used to investigate where signal loss happens within the system.

We anticipate to compare the simulated signals to real data and adjust the simulation to reflect the real data presented. We can then use information gathered to determine what is incorrect about our assumptions and to improve signal collection.

2 Procedure

1. Dr. Kidd and I will initially meet daily to work through the Geant4 installation process and to ensure that the dependencies and requirements are working correctly. We will work through the Geant4 tutorials and build off those to iteratively produce the simulation described above. Once Geant4 is working and I have a more complete understanding of how it works, we will meet two to three times a week to update on progress.
2. Start working on a Geant4 simulation with a radioactive source and a monolithic scintillator. This creates the groundwork of basic interactions and expected data.
3. Replace monolithic scintillator with tubes to emulate the structure of the actual CsI(Tl) scintillator. The new structure simulation will produce more accurate paths and data that reflect the data seen.
4. Light output simulation of microscope either in Geant4 or some other optics simulator. This will simulate the photons as they travel through the microscope body and produce data that can identify causes of loss in light output.

In each step we will compare results with the actual data. I expect much of what is reported will need to be troubleshooted and altered based on outcomes. Then, with the data thus far collected, we will compare images of events with the simulated ones and determine differences with and without the microscope body to determine how much signal is being lost.

References

- [1] M. Kidd. “CAREER: Applying Nuclear Physics to Microbial Ecology”. Unpublished project description. 2014.
- [2] E. Shaw. “Imaging Low Energy Beta Decays Using an EMCCD”. Unpublished report. 2013.
- [3] E. Shaw et al. “SLIMER Long Writeup Version 1.0: Evidence For Observation of ^{14}C ”. Unpublished manuscript. 2013.